Big Data Analytics
CISELEC 501A-CS41S1

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Laboratory Activity #05: Building Classifier Model Using WEKA

Total Points: 130

OBJECTIVE:

I. BUSINESS UNDERSTANDING

A. In this CRISP-DM exercise, your task is to build a classification model that achieves the highest possible accuracy by experimenting with different combinations of data preparation techniques. You'll be expected to apply sanitation, feature selection, and feature extraction methods to improve model performance and compare their effects systematically. As you move through the phases—from Business Understanding to Modeling—consider how each preprocessing choice influences the final outcome. Your goal is not just to build a model, but to demonstrate the value of thoughtful data preparation in predictive analytics.

B. Reflection (10 pts):

1. Based on your understanding of model creation combined with data preparation task(s), do you see any benefit in performing this task? Please elaborate your answer. (**10 pts**)
   a) Yes, there is a benefit from performing this task. Understanding model creation with data preparation helps us greatly in identifying how we can classify and predict our class label. The process where we can eliminate, sanitate, scale, maintain our data, can ensure a tremendous amount of reliable data. By having this reliable data, we can safely execute and even deploy our model creation.

II. DATA UNDERSTANDING

A. Extract the dataset from our Git repository located in the '20251004-Lab-Wk11' directory, and perform your own assessment and evaluation of the file's contents.

B. Reflection (40 pts):

	A	B	C	D	
1	Unique id	channel_name	category	Sub-category	Customer Remarks
2	7e9ae164-6a8b-4521-a2d4-58f7c9fff13f	Outcall	Product Queries	Life Insurance	
3	b07ec1b0-f376-43b6-86df-ec03da3b2e16	Outcall	Product Queries	Product Specific Information	
4	200814dd-27c7-4149-ba2b-bd3af3092880	Inbound	Order Related	Installation/demo	
5	eb0d3e53-c1ca-42d3-8486-e42c8d622135	Inbound	Returns	Reverse Pickup Enquiry	
6	ba903143-1e54-406c-b969-46c52f92e5df	Inbound	Cancellation	Not Needed	
7	1cfde5b9-6112-44fc-8f3b-892196137a62	Email	Returns	Fraudulent User	
8	11a3ffd8-1d6b-4806-b198-c60b5934c9bc	Outcall	Product Queries	Product Specific Information	
9	372b51a5-fa19-4a31-a4b8-a21de117d75e	Inbound	Returns	Exchange / Replacement	Very good
10	6e4413db-4e16-42fc-ac92-2f402e3df03c	Inbound	Returns	Missing	Shopzilla app and it's all coustomer care services is very good service provided all time

1. Based on your assessment, what do you think the dataset is about? (5 pts)
   a) Dataset is about a particular e-commerce customer service performance log. One thing that led me to this is the agent name, supervisor, survey response, and the CSAT score.
2. What are the details of the dataset:
   a) List the field names along with their corresponding data types. (15 pts)
      (1)

Unique id	nominal
channel_name	nominal
category	nominal
Sub-category	nominal
Customer Remarks	nominal
Order_id	nominal
order_date_time	nominal
Issue_reported at	nominal
issue_responded	nominal
Survey_response_Date	nominal

Customer_City	nominal
Product_category	nominal
Item_price	numeric
connected_handling_time	numeric
Agent_name	nominal
Supervisor	nominal
Manager	nominal
Tenure Bucket	nominal
Agent Shift	nominal
CSAT Score	nominal

b) Indicate the number of records. (**2 pts**)
   (1) **85907** number of records
c) Identify the attribute in the dataset that is suitable to serve as the class label. (**3 pts**)
   (1) **CSAT Score Attribute**. It can also be Agent Shift if we want to know what most likely a 5 rating agent picks a certain Shift. hehe

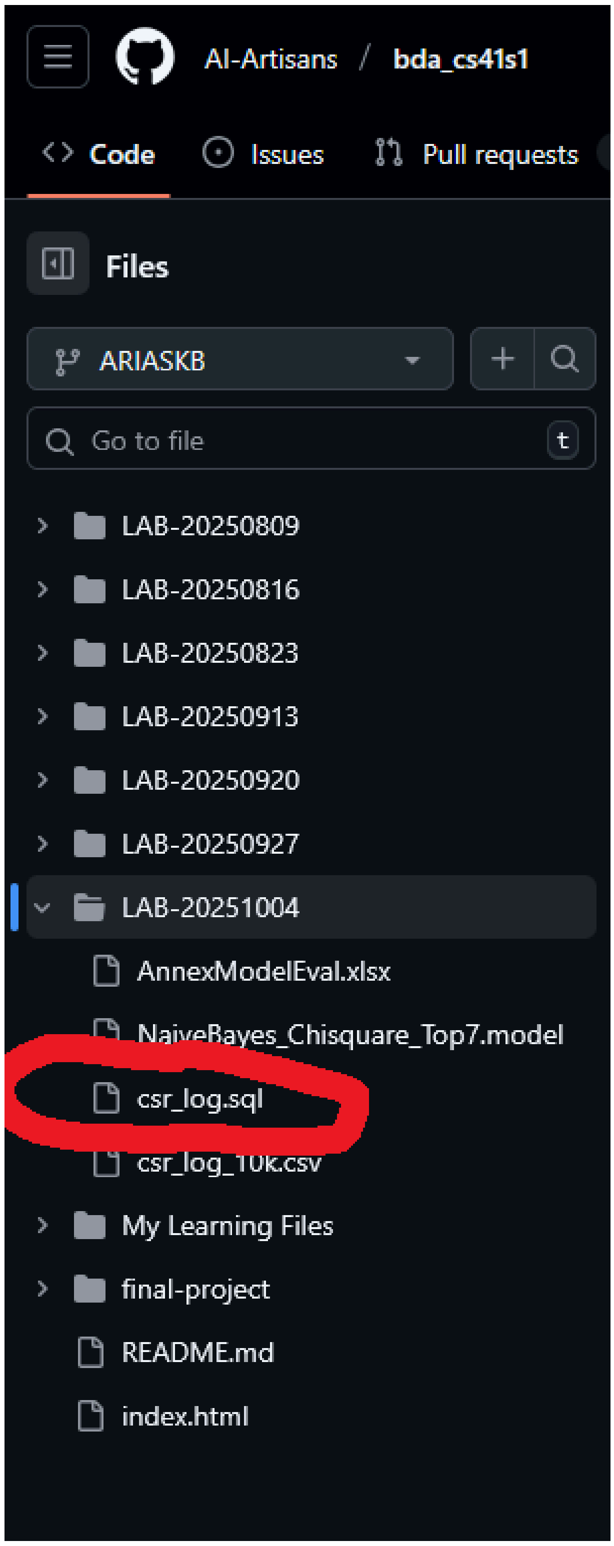
3. Provide a thorough analysis of the dataset in terms of its readiness for model evaluation. Do you observe any anomalies in the dataset? (**15 pts**)
   a) I did observe some anomalies. Those are the **empty fields** that can and are somehow related to other attributes and/or to our label class. We can **try imputing them as '?'** so that we don't have to drop them and can still be read by **WEKA.**
   b) Aside from that, there are also too many instances for an empty field in **Product_category and Customer_City** attributes which can affect the class label. We can **try to impute them as Others and Unknown** as it can be meaningful.
   c) ~~**Quantitative** fields such as **connected_handling_time** and **item_price** should be preserved as they can yield meaningful results such as when CSAT correlates to longer handling time or even higher item_price. (**Try to impute as 0**)~~

III. DATA PREPARATION

A. Your task is to import the dataset into a MySQL DB table.

B. Reflection (40 pts):

1. Provide a detailed description of the process you performed on the dataset during the data preparation stage? Was this necessary based on your observations from the "Data Understanding" stage? (**30 pts**)
   a) Applying not null and unique key to the unique_id attribute. This is necessary as it will ensure there are no other duplicates of a customer report and that it is valid
   b) Apply not null to those fields that are not empty. This hints us that these attributes are all filled and should not be empty at all
   c) **Consider Dropping customer remarks for modelling** unless we have to do an NLP task for considering a semantic meaning to those remarks. **But keep them on MySQL DB for future use.**
   d) Impute other attributes that have empty fields to be marked as a meaningful instance.
      (1) **Product_category**. **Impute fields as Others**. Can be highly meaningful as it is divided into factions rather than unique identifiers.
      (2) **Customer_City**. **Impute fields as Unknown**. Can be meaningful and might correlate to our class label.
      (3) Make sure to treat **CSAT Score** as **nominal** instead of numeric as CSAT score seems to be an **ordinal** here. It has a ranking.
      ~~(4) Impute 0 to connected_handling_time and item_price~~
   e) Replace remaining empty fields as '?' so that weka can read the export
2. Upload your MySQL DB table dump to your own Git branch. (**10 pts**)

a) 
AI-Artisans / bda_cs41s1
Code
Issues
Pull requests
Files
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t
LAB-20250809
LAB-20250816
LAB-20250823
LAB-20250913
LAB-20250920
LAB-20250927
LAB-20251004
AnnexModelEval.xlsx
NaiveBayes_Chisquare_Top7.model
csr_log.sql
csr_log_10k.csv
My Learning Files
final-project
README.md
index.html

IV. MODELING

A. Your task is to select three (3) different classification algorithms to perform modeling that yields the highest accuracy during the testing activity.

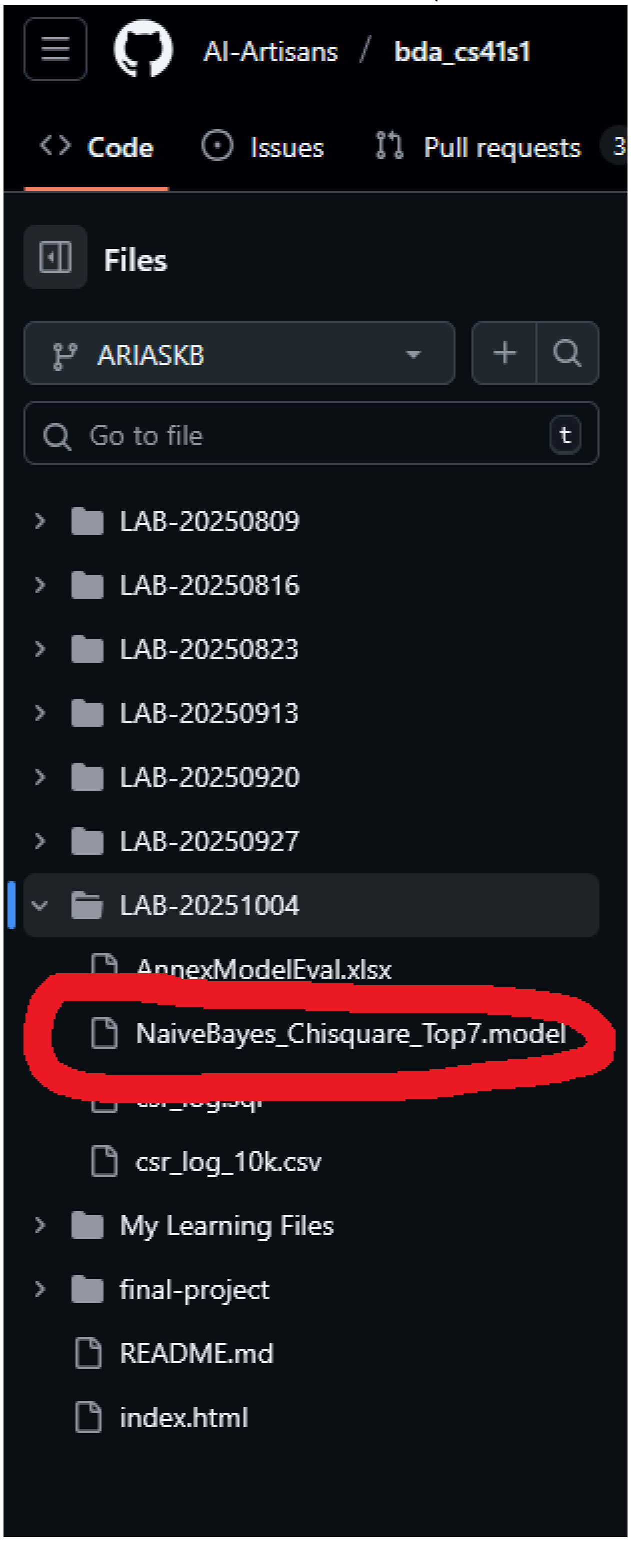
B. Reflection (40 pts):

1. Using your top three (3) classification algorithms, complete the table below with your recorded measurements (30 pts):

Evaluators	**ReliefF**												**CfsSubset (GreedyWise w/ Rank)**			**Chi Squared or InfoGain**		
Classifiers	**Trees >> J48**				**Bayes >> NaïveBayes**				**Functions >> LibSVM**				**Bayes >> NaïveBayes**			**Bayes >> NaïveBayes**		
Attributes	All	Top 14	Top 10	**Top 7**	All	Top 14	Top 10	Top 7	**All**	Top 14	Top 10	Top 7	Top 14	Top 10	Top 7	Top 14	Top 10	**Top 7**
Correctly Classified Instances	68.76	68.76	68.76	**68.76**	96.92	86.86	70.21	69.83	**69.48**	68.76	68.76	68.76	97.03	96.78	96.39	97.24	97.66	**97.74**
Incorrectly Classified Instances	31.24	31.24	31.24	**31.24**	3.08	13.14	29.79	30.17	**30.52**	31.24	31.24	31.24	2.97	3.22	3.61	2.76	2.34	**2.26**
Kappa statistics	0	0	0	**0**	0.936	0.701	18.53	0.1615	**0.047**	0	0	0	0.9382	0.9321	0.9236	0.9425	0.9511	**0.9527**
Precision (Class: 1)	?	?	?	**?**	0.936	0.819	0.501	0.482	**0.675**	?	?	?	0.946	0.969	0.963	0.957	0.976	**0.984**
Precision (Class: 2)	?	?	?	**?**	0.986	0.857	0.222	0.222	**1**	?	?	?	0.982	1	1	0.959	1	**1**
Precision (Class: 3)	?	?	?	**?**	0.991	0.949	0.286	0.414	**1**	?	?	?	0.996	1	1	0.996	1	**1**
Precision (Class: 4)	?	?	?	**?**	0.986	0.947	0.497	0.494	**0.75**	?	?	?	0.986	0.962	0.957	0.972	0.973	**0.982**
Precision (Class: 5)	0.688	0.688	0.688	**0.688**	0.971	0.865	0.724	0.719	**0.695**	0.688	0.688	0.688	0.971	0.962	0.957	0.972	0.973	**0.982**
Recall (Class: 1)	?	?	0	**0**	0.953	0.715	0.274	0.234	**0.059**	0	0	0	0.962	0.97	0.963	0.955	0.969	**0.98**
Recall (Class: 2)	?	?	0	**0**	0.518	0.17	0.014	0.014	**0.007**	0	0	0	0.397	0.234	0.213	0.504	0.376	**0.291**

Recall (Class: 3)	?	?	0	**0**	0.725	0.301	0.032	0.039	**0.01**	0	0	0	0.735	0.608	0.55	0.731	0.764	**0.731**
Recall (Class: 4)	?	?	0	**0**	0.981	0.69	0.066	0.059	**0.011**	0	0	0	0.982	0.983	0.978	0.983	0.987	**0.988**
Recall (Class: 5)	1	1	1	**1**	0.99	0.973	0.953	0.956	**0.996**	1	1	1	0.992	0.996	0.995	0.994	0.998	**1**
F-Measures (Class: 1)	?	?	?	**0**	0.945	0.763	0.354	0.315	**0.108**	?	0	0	0.954	0.97	0.963	0.956	0.973	**0.982**
F-Measures (Class: 2)	?	?	?	**0**	0.679	0.284	0.027	0.027	**0.014**	?	0	0	0.566	0.379	0.351	0.66	0.546	**0.451**
F-Measures (Class: 3)	?	?	?	**0**	0.837	0.457	0.058	0.071	**0.019**	?	0	0	0.845	0.757	0.71	0.843	0.866	**0.845**
F-Measures (Class: 4)	?	?	?	**0**	0.984	0.798	0.116	0.106	**0.022**	?	0	0	0.984	0.988	0.988	0.985	0.988	**0.988**
F-Measures (Class: 5)	0.815	0.815	0.815	**0.815**	0.981	0.916	0.823	0.821	**0.819**	0.815	0.815	0.815	0.981	0.978	0.976	0.983	0.985	**0.986**

2. Upload the model that achieves the highest measurement to your Git branch. (**10 pts**)
NOTE: To build your model, follow these steps:
   a) In the "WEKA Explorer" window, under the "Result list" section, select the recorded result with the highest evaluation and right-click.
   b) From the options, select "Save model".
   c) In the "Save" window, choose the destination where you want to save the model file (with the .model extension)



" I affirm that I have not given or received any unauthorized help in this assignment, and that this work is my own "

Kenneth Arias